Figure 2. Nauru: Public Sector Debt Sustainability Analysis (DSA) – Baseline Scenario

(In percentage of GDP otherwise indicated)

Debt, Economic and Market Indicators [1/]

	Actual			Projections					
	2011-2016 [2/]	2017	2018	2019	2020	2021	2022	2023	2024
Nominal gross public debt	141.9	90.4	86.8	78.4	75.7	73.6	71.5	68.9	79.1
Of which: guarantees	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Public gross financing needs	-3.7	-21.4	-30.9	-1.4	-14.3	-3.7	-2.1	-0.5	1.5
Real GDP growth (in percent)	14.3	-5.5	5.7	1.0	0.6	1.1	1.5	1.8	2.0
Inflation (GDP deflator, in percent)	4.6	11.9	4.1	2.3	2.3	1.9	1.6	2.0	2.0
Nominal GDP growth (in percent)	18.2	5.7	10.1	3.3	2.9	3.0	3.1	3.9	4.1
Effective interest rate (in percent) [4/]	0.7	0.0	0.2	0.0	0.1	0.1	0.1	0.1	0.1

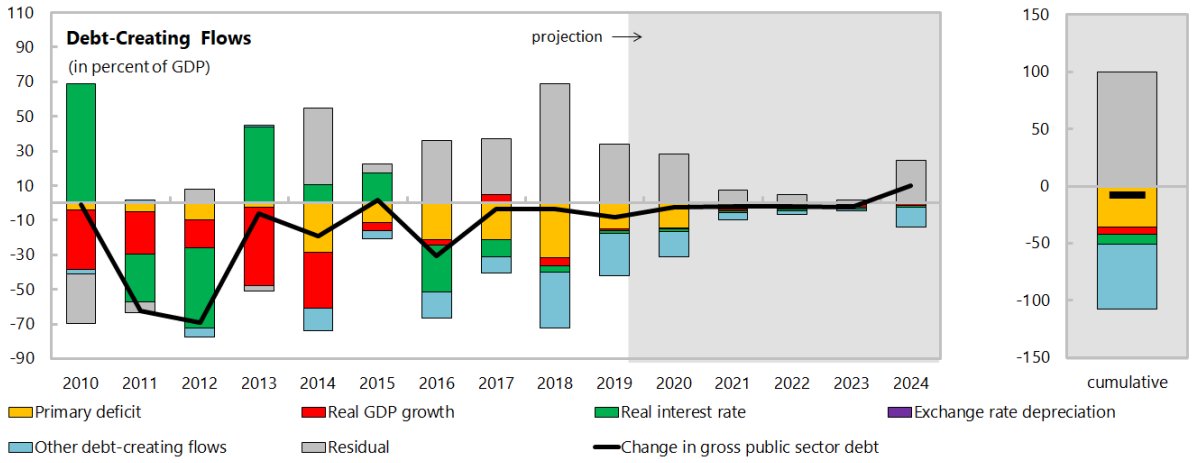
As of June 13, 2019

Sovereign Spreads	
EMBIG (bp) 3/	n.a.
5Y CDS (bp)	n.a.

Ratings	Foreign	Local
Moody's	n.a.	n.a.
S&Ps	n.a.	n.a.
Fitch	n.a.	n.a.

Contribution to Changes in Public Debt

	Actual			Projections							
	2011-2016	2017	2018	2019	2020	2021	2022	2023	2024	cumulative	debt-stabilizing primary balance [9/]
Change in gross public sector debt	-31.0	-3.6	-3.6	-8.4	-2.7	-2.1	-2.1	-2.5	10.2	-7.6	
Identified debt-creating flows	-45.0	-35.8	-72.5	-42.3	-31.0	-9.7	-6.8	-4.4	-13.5	-107.7	
Primary deficit	-13.1	-21.2	-31.9	-15.0	-14.5	-3.7	-2.4	-1.2	0.7	-36.2	-14.2
Primary (noninterest) revenue and grants	77.0	121.7	128.5	140.6	134.9	102.5	100.6	99.6	97.1	675.4	
Primary (noninterest) expenditure	63.8	100.5	96.6	125.6	120.4	98.8	98.2	98.4	97.8	639.2	
Automatic debt dynamics [5/]	-25.9	-5.1	-8.1	-2.8	-2.2	-2.2	-2.2	-2.6	-2.6	-14.5	
Interest rate/growth differential [6/]	-25.9	-5.1	-8.1	-2.8	-2.2	-2.2	-2.2	-2.6	-2.6	-14.5	
Of which: real interest rate	-5.0	-10.0	-3.4	-2.0	-1.7	-1.3	-1.1	-1.3	-1.3	-8.7	
Of which: real GDP growth	-20.9	4.9	-4.7	-0.8	-0.5	-0.8	-1.1	-1.3	-1.4	-5.8	
Exchange rate depreciation [7/]	0.0	0.0	0.0	...	...	...	...	...	...	...	
Other identified debt-creating flows	-6.0	-9.5	-32.5	-24.5	-14.3	-3.8	-2.2	-0.7	-11.5	-57.0	
Fiscal: Net acquisition of financial assets (negative)	-6.0	-9.5	-32.5	-24.5	-14.3	-3.8	-2.2	-0.7	-11.5	-57.0	
Contingent liabilities	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	
Please specify (2) (e.g., ESM and Euroarea loans)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	
Residual, including asset changes [8/]	14.0	32.2	68.9	34.0	28.3	7.6	4.7	1.9	23.7	100.1	



Source: IMF staff.

1/ Public sector is defined as general government and includes public guarantees, defined as Arrears.

2/ Based on available data.

3/ Long-term bond spread over German bonds.

4/ Defined as interest payments divided by debt stock (excluding guarantees) at the end of previous year.

5/ Derived as $[(r - \pi(1+g) - g + ae(1+r)]/(1+g+\pi+g\pi))$ times previous period debt ratio, with r = interest rate; π = growth rate of GDP deflator; g = real GDP growth rate; a = share of foreign-currency denominated debt; and e = nominal exchange rate depreciation (measured by increase in local currency value of U.S. dollar).

6/ The real interest rate contribution is derived from the numerator in footnote 5 as $r - \pi(1+g)$ and the real growth contribution as -g.

7/ The exchange rate contribution is derived from the numerator in footnote 5 as $ae(1+r)$.

8/ Includes changes in the stock of guarantees, asset changes, and interest revenues (if any). For projections, includes exchange rate changes during the projection period.

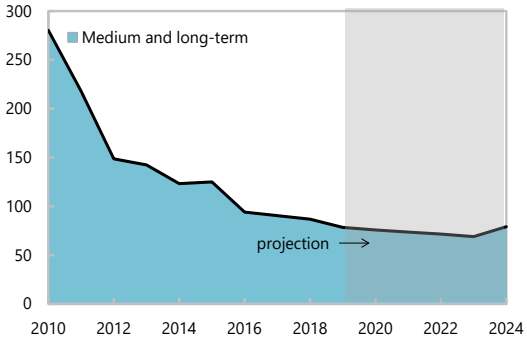
9/ Assumes that key variables (real GDP growth, real interest rate, and other identified debt-creating flows) remain at the level of the last projection year.

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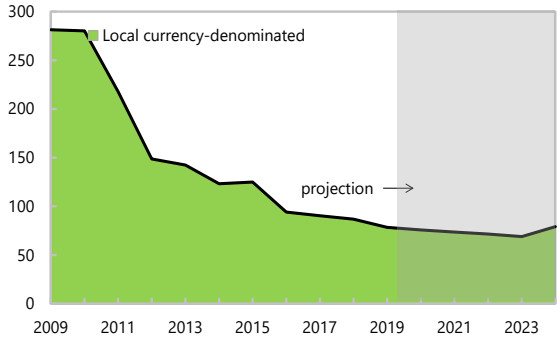
Figure 3. Nauru: Public DSA – Composition of Public Debt and Alternative Scenarios

Composition of Public Debt

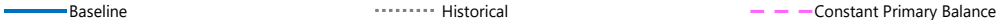
By Maturity
(in percent of GDP)



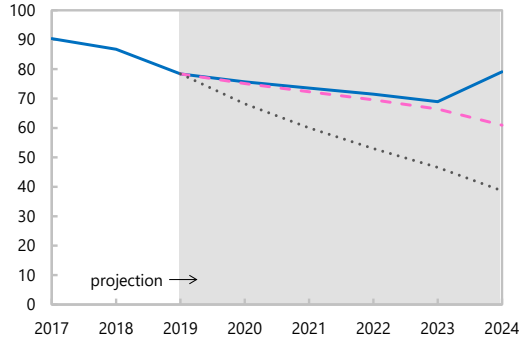
By Currency
(in percent of GDP)



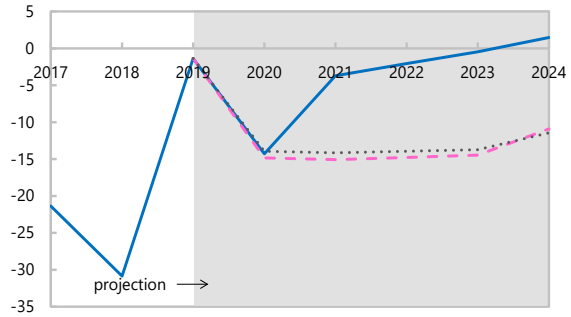
Alternative Scenarios



Gross Nominal Public Debt
(in percent of GDP)



Public Gross Financing Needs
(in percent of GDP)



Underlying Assumptions
(in percent)

Baseline Scenario	2019	2020	2021	2022	2023	2024
Real GDP growth	1.0	0.6	1.1	1.5	1.8	2.0
Inflation	2.3	2.3	1.9	1.6	2.0	2.0
Primary Balance	15.0	14.5	3.7	2.4	1.2	-0.7
Effective interest rate	0.0	0.1	0.1	0.1	0.1	0.1
Constant Primary Balance Scenario						
Real GDP growth	1.0	0.6	1.1	1.5	1.8	2.0
Inflation	2.3	2.3	1.9	1.6	2.0	2.0
Primary Balance	15.0	15.0	15.0	15.0	15.0	15.0
Effective interest rate	0.0	0.1	0.1	0.1	0.1	0.1

Historical Scenario	2019	2020	2021	2022	2023	2024
Real GDP growth	1.0	10.6	10.6	10.6	10.6	10.6
Inflation	2.3	2.3	1.9	1.6	2.0	2.0
Primary Balance	15.0	14.1	14.1	14.1	14.1	14.1
Effective interest rate	0.0	0.1	0.1	0.1	0.1	0.1

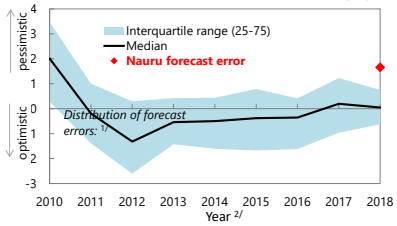
Source: IMF staff.

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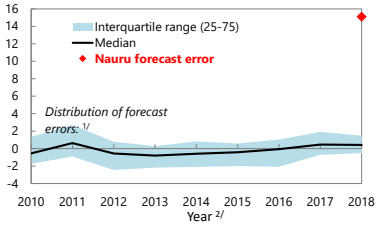
Figure 4. Nauru: Public DSA – Realism of Baseline Assumptions

Forecast Track Record, versus all countries

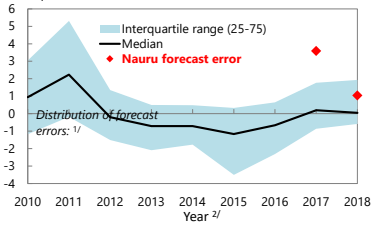
Real GDP Growth
(in percent, actual-projection)
Nauru median forecast error, 2010-2018: 1.66
Has a percentile rank of: 94%



Primary Balance
(in percent of GDP, actual-projection)
Nauru median forecast error, 2010-2018: 15.10
Has a percentile rank of: 100%

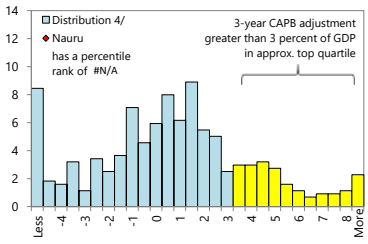


Inflation (Deflator)
(in percent, actual-projection)
Nauru median forecast error, 2010-2018: 2.32
Has a percentile rank of: 91%

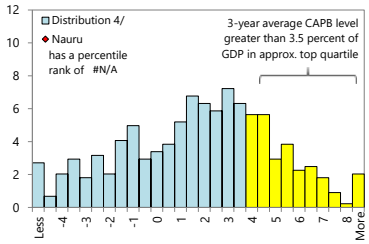


Assessing the Realism of Projected Fiscal Adjustment

3-Year Adjustment in Cyclically-Adjusted Primary Balance (CAPB)
(Percent of GDP)

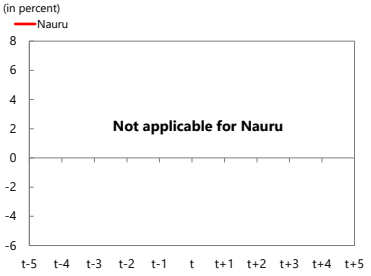


3-Year Average Level of Cyclically-Adjusted Primary Balance (CAPB)
(Percent of GDP)



Boom-Bust Analysis 3/

Real GDP growth
(in percent)

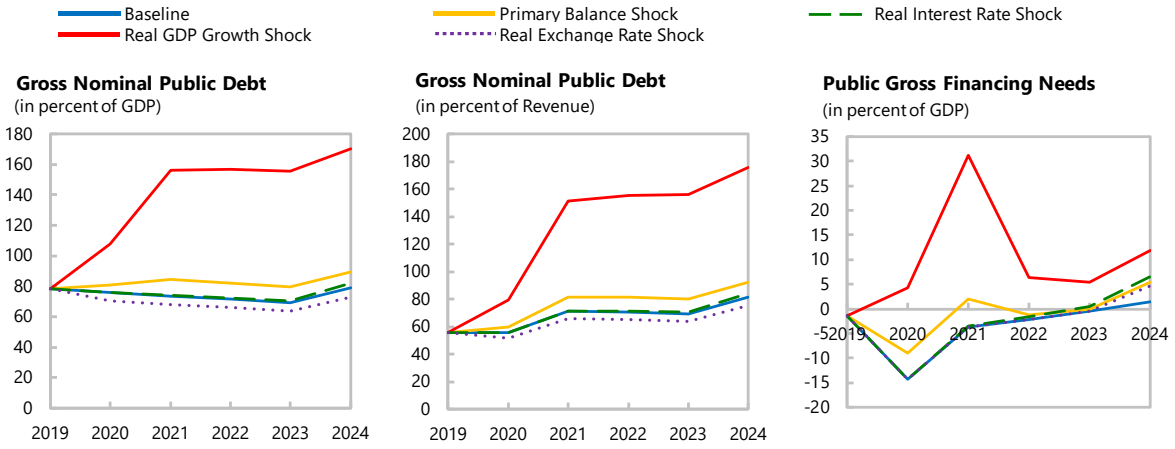


Source : IMF Staff.
1/ Plotted distribution includes all countries, percentile rank refers to all countries.
2/ Projections made in the spring WEO vintage of the preceding year.
3/ Not applicable for Nauru, as it meets neither the positive output gap criterion nor the private credit growth criterion.
4/ Data cover annual obervations from 1990 to 2011 for advanced and emerging economies with debt greater than 60 percent of GDP. Percent of sample on vertical axis.

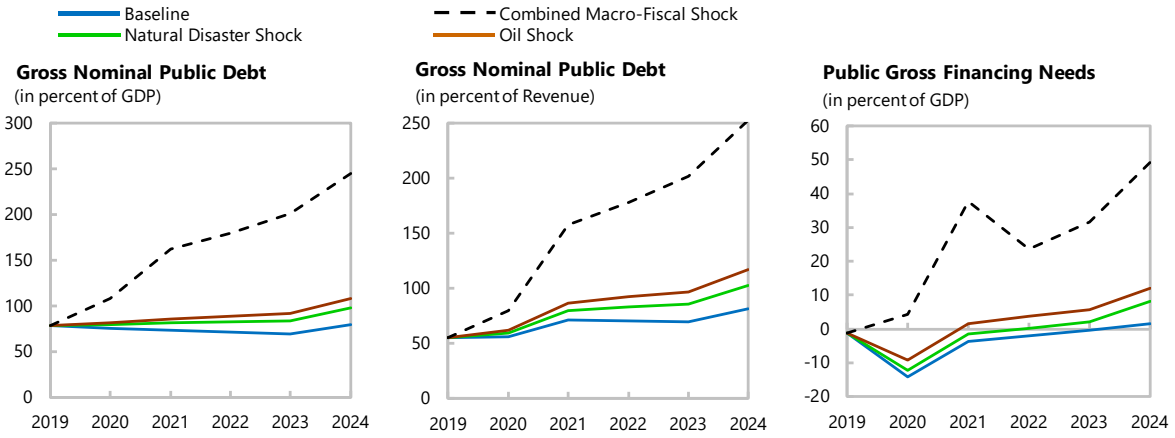
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Figure 5. Nauru: Public DSA – Stress Tests

Macro-Fiscal Stress Tests



Additional Stress Tests



Underlying Assumptions

(in percent)

Primary Balance Shock	2019	2020	2021	2022	2023	2024
Real GDP growth	1.0	0.6	1.1	1.5	1.8	2.0
Inflation	2.3	2.3	1.9	1.6	2.0	2.0
Primary balance	15.0	9.1	-1.7	2.4	1.2	-0.7
Effective interest rate	0.0	0.1	0.3	0.4	0.4	0.4
Real Interest Rate Shock						
Real GDP growth	1.0	0.6	1.1	1.5	1.8	2.0
Inflation	2.3	2.3	1.9	1.6	2.0	2.0
Primary balance	15.0	14.5	3.7	2.4	1.2	-0.7
Effective interest rate	0.0	0.1	0.4	0.8	1.3	2.0
Combined Shock						
Real GDP growth	1.0	-10.4	-9.9	1.5	1.8	2.0
Inflation	2.3	-0.4	-0.8	1.6	2.0	2.0
Primary balance	15.0	-4.0	-28.9	2.4	1.2	-0.7
Effective interest rate	0.0	0.1	6.6	13.4	15.9	18.3

Real GDP Growth Shock	2019	2020	2021	2022	2023	2024
Real GDP growth	1.0	-10.4	-9.9	1.5	1.8	2.0
Inflation	2.3	-0.4	-0.8	1.6	2.0	2.0
Primary balance	15.0	-4.0	-28.9	2.4	1.2	-0.7
Effective interest rate	0.0	0.1	1.2	3.0	2.8	2.7
Real Exchange Rate Shock						
Real GDP growth	1.0	0.6	1.1	1.5	1.8	2.0
Inflation	2.3	9.4	1.9	1.6	2.0	2.0
Primary balance	15.0	14.5	3.7	2.4	1.2	-0.7
Effective interest rate	0.0	0.1	0.1	0.1	0.1	0.1

Source IMF staff.

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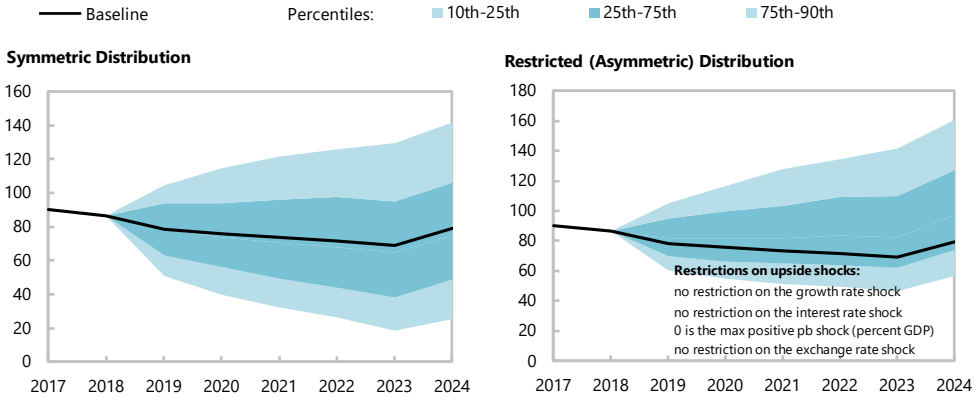
Figure 6. Nauru: Public DSA Risk Assessment

Heat Map

Debt level [1/]	Real GDP Growth Shock	Primary Balance Shock	Real Interest Rate Shock	Exchange Rate Shock	Contingent Liability shock
Gross financing needs [2/]	Real GDP Growth Shock	Primary Balance Shock	Real Interest Rate Shock	Exchange Rate Shock	Contingent Liability Shock
Debt profile [3/]	Market Perception	External Financing Requirements	Change in the Share of Short Term Debt	Public Debt Held by Non-Residents	Foreign Currency Debt

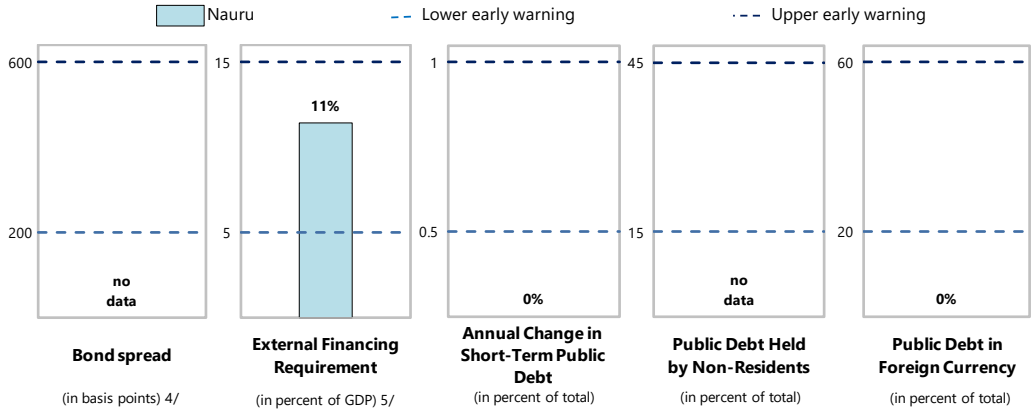
Evolution of Predictive Densities of Gross Nominal Public Debt

(in percent of GDP)



Debt Profile Vulnerabilities

(Indicators vis-à-vis risk assessment benchmarks, in 2018)



Source: IMF staff.

1/ The cell is highlighted in green if debt burden benchmark of 70% is not exceeded under the specific shock or baseline, yellow if exceeded under specific shock but not baseline, red if benchmark is exceeded under baseline, white if stress test is not relevant.

2/ The cell is highlighted in green if gross financing needs benchmark of 15% is not exceeded under the specific shock or baseline, yellow if exceeded under specific shock but not baseline, red if benchmark is exceeded under baseline, white if stress test is not relevant.

3/ The cell is highlighted in green if country value is less than the lower risk-assessment benchmark, red if country value exceeds the upper risk-assessment benchmark, yellow if country value is between the lower and upper risk-assessment benchmarks. If data are unavailable or indicator is not relevant, cell is white.
Lower and upper risk-assessment benchmarks are:

200 and 600 basis points for bond spreads; 5 and 15 percent of GDP for external financing requirement; 0.5 and 1 percent for change in the share of short-term debt; 15 and 45 percent for the public debt held by non-residents; and 20 and 60 percent for the share of foreign-currency denominated debt.

4/ Long-term bond spread over German bonds, an average over the last 3 months, .

5/ External financing requirement is defined as the sum of current account deficit, amortization of medium and long-term total external debt, and short-term total external debt at the end of previous period.

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